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Date	09-02-23	Version 02	Edge Marks	N/A		
Description	Synflex 275mg 550mg 100 tabs		Min. Font Size	8 pt	Min. Code Size	7 pt
Pharmanovia Previous Code		738-8523	Notes			
				Non Printable Colours		
				Guides		

PACK INSERT FOR MALAYSIA

Synflex[®]

ATNAHS

Naproxen sodium

1. DESCRIPTION
1.1 Therapeutic / Pharmacologic Class of Drug
Non-steroidal anti-inflammatory agent.
1.2 Type of DosageForm
Synflex Tablets:
275 mg: Light blue, oval film-coated tablet, with marking NPS 275 on one side.
550 mg: Dark blue, oblong film-coated tablet, with marking NPS 550 on one side.
1.3 Route of Administration
Oral
1.4 Sterile / Radioactive Statement
Not applicable
1.5 Qualitative and Quantitative Composition
Active ingredient: naproxen sodium.
Synflex Tablets: film-coated tablets containing naproxen sodium 275 mg, 550 mg Naproxen sodium is a white to creamy white, crystalline solid, freely soluble in water at neutral pH.
List of Excipients
Microcrystalline cellulose, povidone K29/32, talc, magnesium stearate, purified water, opadry (YS-1-4215/16).

2. CLINICAL PARTICULARS
2.1 Therapeutic Indication(s)
Synflex (naproxen sodium) is indicated in the relief of mild to moderate pain including post partum pain, pain following IUD insertion, post-operative pain and pain due to orthopedic surgery, for the treatment of primary dysmenorrhea and for the relief (prophylaxis) of migraine headache. It is also indicated for the treatment of the signs and symptoms of mild to moderately severe, acute or chronic, musculoskeletal and soft tissue inflammation and acute gout.
2.2 Dosage and Administration
General
After assessing the risk/benefit ratio in each individual patient, the lowest effective dose for the shortest possible duration should be used. Although naproxen and naproxen-sodium-containing products all circulate in the plasma as naproxen, they have pharmacokinetic differences that may affect onset of action. Onset of pain relief can begin within 30 minutes in patients taking Synflex and within 1 hour in patients taking Naprosyn.
The recommended strategy for initiating therapy is to choose a formulation and a starting dose likely to be effective for the patient and then adjust the dosage based on observation of benefit and/or adverse events.
A lower dose should be considered in patients with renal or hepatic impairment or in elderly patients. Naprosyn/Synflex is not recommended in patients with baseline creatinine clearance less than 30 ml/minute because accumulation of naproxen metabolites has been seen in patients with severe renal failure or those on dialysis.
Recommended formulations
Because the sodium salt of naproxen is more rapidly absorbed, Synflex is recommended for the management of acute painful conditions when prompt onset of pain relief is desired. Synflex may be given orally either in fasting state or with meals and/or antacids.
Dose in adults
Acute conditions
Analgesia / Dysmenorrhea / Acute musculoskeletal conditions/Acute pain states in which there is an inflammatory component.
Because the sodium salt of naproxen is more rapidly absorbed, Synflex is recommended for the management of acute painful conditions when prompt onset of pain relief is desired.
The recommended starting dose is Synflex 550 mg followed by Synflex 275 mg every 6-8 hours as required.
Acute gout: The recommended starting dose is 825 mg of Synflex followed by 275 mg every 8 hours as needed.
Migraine: For treatment of acute migraine headache, the dose is Synflex 825 mg at the first symptom of an impending attack. An additional dose of Synflex 275 mg to 550 mg or Naprosyn 250 mg to 500 mg can be taken throughout the day, if necessary, but not before half an hour after the initial dose.
For prophylaxis of migraine headache, the dose of Synflex is 550 mg twice daily. If no improvement is seen within 4 to 6 weeks, the drug should be discontinued. A total daily dose of 1375mg per day should not be exceeded.
Dose in children
Synflex is not recommended for use in children under 16 years of age.
2.3 Contraindications
Naprosyn/Synflex are contraindicated in patients who have had allergic reactions to prescription as well as to over-the-counter products containing naproxen or naproxen sodium. They are also contraindicated in patients in whom aspirin or other nonsteroidal anti-inflammatory/ analgesic drugs induce the syndrome of asthma, rhinitis and nasal polyps. Both types of reactions have the potential of being fatal. Severe anaphylactic-like reactions to naproxen have been reported in such patients. Naprosyn/Synflex are contraindicated in patients with active or a history of gastrointestinal bleeding or perforation, related to previous NSAIDs therapy. Active, or history of recurrent peptic ulcer/ hemorrhage (two or more distinct episodes of proven ulceration or bleeding). As with other NSAIDs, Naprosyn/Synflex are contraindicated in patients with severe heart failure. Naprosyn/Synflex are contraindicated in children under 2 years of age since safety in this age group has not been established. Naprosyn/Synflex are contraindicated in patients with severe liver impairment (in line with Child-Pugh classification C), examples of which include severe hepatitis and liver failure. Naprosyn/Synflex are contraindicated in patients with severe renal impairment (GFR <30 ml/min).

WARNINGS RISK OF GI ULCERATION, BLEEDING AND PERFORATION WITH NSAIDS
Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without symptoms, in patients treated with NSAID therapy. Although minor GI problems (e.g. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms. Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious adverse events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

2.4 Warnings and Precautions
2.4.1 General
The use of Naprosyn/Synflex with concomitant NSAIDs including cyclooxygenase-2 selective inhibitors should be avoided. Undesirable effects may be minimized by using the lowest effective dose for the shortest duration necessary to control symptoms (see Section 2.2, Dosage and Administration, and Gastrointestinal effects and Cardiovascular and cerebrovascular effects, below).
Gastrointestinal ulceration, bleeding and perforation GI bleeding, ulceration or perforation, which can be fatal, has been reported with all NSAIDs, including Naprosyn/Synflex therapy at any time during treatment, with or without warning symptoms or a previous history of serious GI events. Studies to date have not identified any subset of patients not at risk of developing peptic ulcer and bleeding. The elderly have an increased frequency of adverse reactions, to NSAIDs, especially gastrointestinal bleeding and perforation which may be fatal. Debilitated patients seem to tolerate ulceration or bleeding less well than others. Most of the fatal gastrointestinal events associated with non-steroidal anti-inflammatory drugs occurred in the elderly and/or debilitated patients. The risk of GI bleeding, ulceration or perforation is higher with increasing NSAID doses, in patients with a history of ulcer, particularly if complicated with haemorrhage or perforation (See section 2.3 contraindications) and in the elderly. These patients should commence treatment on the lowest dose available. Combination therapy with protective agents (e.g. misoprostol or proton pump inhibitors) should be considered for these patients, and also for patients requiring concomitant low dose aspirin or other drugs likely to increase gastrointestinal risk (see below and Section 2.4.3 Interactions with other Medicinal Products and Other forms of Interaction)

NSAIDs should be given with care to patients with a history of inflammatory bowel disease (ulcerative colitis, Crohn's disease) as their condition may be exacerbated (See section 2.6 Undesirable Effects) Patients with a history of GI toxicity, particularly when elderly, should report any unusual abdominal symptoms (especially GI bleeding) particularly in the initial stages of treatment. When GI bleeding or ulceration occurs in patients receiving Naprosyn/Synflex, treatment should be withdrawn. In patients with a history of gastrointestinal disease, Naprosyn/Synflex should be given under close supervision. Open studies in patients with rheumatoid arthritis who had upper gastrointestinal dysfunction and/or were intolerant of other commonly used NSAIDs indicated that Naprosyn/ Synflex is generally well tolerated. In clinical trials, the number of patients receiving Naprosyn EC who developed severe gastrointestinal symptoms was decreased as compared to patients who received Naprosyn. The number of patients who withdrew prematurely for gastrointestinal symptoms was reduced among those who received Naprosyn EC. As with other non-steroidal anti-inflammatory drugs, the incidence and severity of gastrointestinal complications may increase with increasing dose and duration of treatment with Naprosyn/Synflex. Caution should be advised in patients receiving concomitant medications which could increase the risk of ulceration or bleeding, such as oral corticosteroids, anticoagulants such as warfarin, selective serotonin-reuptake inhibitors or anti-platelet agents such as aspirin (See section 2.4.3, Interactions with other Medicinal Products and other Forms of Interaction)
Precautions related to elderly patients
The elderly have an increased frequency of adverse reactions to NSAIDs especially gastrointestinal bleeding and perforation which may be fatal. In elderly patients the clearance is reduced. Use of the lower end of the dosage range is recommended (see section 2.2 Dosage and Administration).
Skin reactions
Serious skin reactions, some of them fatal, including exfoliative dermatitis, Stevens-Johnson syndrome, Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) and toxic epidermal necrolysis, have been reported very rarely in association with the use of NSAIDs (See section 2.6, Undesirable Effects). Patients appear to be at highest risk of these reactions early in the course of therapy, the onset of the reaction occurring in the majority of cases within the first month of treatment. Naprosyn/Synflex should be discontinued at the first appearance of skin rash, mucosal lesions, or any other sign of hypersensitivity.
Anaphylactic (anaphylactoid) reactions
Hypersensitivity reactions may occur in susceptible individuals. Anaphylactic (anaphylactoid) reactions may occur both in patients with and without a history of hypersensitivity or exposure to aspirin, other non-steroidal anti-inflammatory drugs or naproxen-containing products. They may also occur in individuals with a history of angioedema, bronchospastic reactivity (e.g. asthma), rhinitis and nasal polyps. Anaphylactoid reactions, like anaphylaxis, may have a fatal outcome. Bronchospasm may be precipitated in patients suffering from, or with a history of, asthma or allergic disease or aspirin sensitivity.
Renal effects
There have been reports of impaired renal function, renal failure, acute interstitial nephritis, hematuria, proteinuria, renal papillary necrosis and occasionally nephrotic syndrome associated with Naprosyn/Synflex. As with other NSAIDs, Naprosyn/Synflex should be used with caution in patients with impaired renal function or a history of kidney disease because naproxen is an inhibitor of prostaglandin synthesis. Caution should be observed in patients with conditions leading to a reduction in blood volume and/or renal blood flow as renal prostaglandins have a supportive role in the maintenance of renal perfusion. In these patients administration of Naprosyn/Synflex or other NSAIDs may cause a dose-dependent reduction in renal prostaglandin formation and may precipitate overt renal decompensation or failure. Patients at greatest risk of this reaction are those with impaired renal function hypovolemia, heart failure, liver dysfunction, salt depletion, those taking diuretics, angiotensin converting enzyme inhibitors or angiotensin receptor blockers and the elderly. Discontinuation of Naprosyn/Synflex is usually followed by recovery to the pretreatment state. Naprosyn/Synflex should be used with great caution in such patients and the monitoring of serum creatinine and/or creatinine clearance is advised and patients should be adequately hydrated. A reduction in daily dosage should be considered to avoid the possibility of excessive accumulation of naproxen metabolites in these patients. Naprosyn/Synflex are not recommended in patients with baseline creatinine clearance less than 30 ml/ min because accumulation of naproxen metabolites has been seen in such patients. Hemodialysis does not decrease the plasma concentration of naproxen because of the high degree of its protein binding.
Hepatic effects
As with other non-steroidal anti-inflammatory drugs, elevations of one or more liver function tests may occur. Hepatic abnormalities may be the result of hypersensitivity rather than direct toxicity. Severe hepatic reactions, including jaundice and hepatitis (some cases of hepatitis have been fatal) have been reported with this drug as with other non-steroidal anti-inflammatory drugs. Cross reactivity has been reported.
Hematological
Naproxen decreases platelet aggregation and prolongs bleeding time. This effect should be kept in mind when bleeding times are determined. Patients who have coagulation disorders or are receiving drug therapy that interferes with hemostasis should be carefully observed if Naprosyn/Synflex are administered. Patients at high risk of bleeding and those on full anticoagulation therapy (e.g. dicoumarol derivatives) may be at increased risk of bleeding if given Naprosyn/Synflex concurrently.
Antipyretic effects
The antipyretic and anti-inflammatory activities of naproxen may reduce fever and inflammation, thus diminishing their utility as diagnostic signs.
Steroids
If steroid dosage is reduced or eliminated during therapy, the steroid dosage should be reduced slowly and the patients must be observed closely for any evidence of adverse effects, including adrenal insufficiency and exacerbation of symptoms of arthritis.
Ocular effects
Studies have not shown changes in the eye attributable to Naprosyn/Synflex administration. In rare cases, adverse ocular disorders including papillitis, retrobulbar optic neuritis and papilloedema, have been reported in users of NSAIDs including Naprosyn/Synflex, although a cause-and-effect relationship cannot be established; accordingly, patients who develop visual disturbances during treatment with Naprosyn/ Synflex should have an ophthalmological examination.
Sodium
A 275 mg tablet of Synflex contains approximately 25 mg (about 1 mEq) of sodium, so the total amount of sodium ingested with the maximum recommended daily dose is 125 mg, about 16% of the 800 mg of sodium permitted on a severely sodium-restricted diet. This should be considered in patients whose overall intake of sodium should be markedly restricted.
Sodium/fluid retention in cardiovascular conditions and peripheral edema
Although sodium retention has not been reported in metabolic studies with Naprosyn/Synflex, patients with compromised cardiac function may be at greater risk when taking Naprosyn/Synflex, and caution is required in patients with a history of hypertension and /or heart failure as fluid retention and edema have been reported in association with NSAID therapy. Peripheral edema has been observed in some patients taking Naprosyn/Synflex or other NSAIDs.
Cardiovascular and cerebrovascular events
Appropriate monitoring and advice are required for patients with a history of hypertension and/ or mild to moderate congestive heart failure as fluid retention and edema have been reported in association with NSAID therapy. Clinical trial and epidemiological data suggest that use of coxibs and some NSAIDs (particularly at high doses and long term treatment) may be associated with a small increased risk of arterial thrombotic events (for example myocardial infarction or stroke). Although data suggest that the use of naproxen (1000 mg/d) may be associated with a lower risk, some risk cannot be excluded. Patients with uncontrolled hypertension, congestive heart failure, established ischemic heart disease, peripheral arterial disease, and/or cerebrovascular disease should only be treated with naproxen after careful consideration. Similar consideration should be made before initiating longer-term treatment of patients with risk factors for cardiovascular disease (e.g. hypertension, hyperlipidaemia, diabetes mellitus, smoking).
Precautions related to fertility
The use of Naprosyn/Synflex, as with any drug

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known to inhibit cyclooxygenase/prostaglandin synthesis, may impair fertility and is not recommended in women attempting to conceive. In women who have difficulty conceiving or are undergoing investigation of infertility, withdrawal of Naprosyn/Synflex should be considered.

Combination with other NSAIDs

The combination of Naprosyn/Synflex and other NSAIDs is not recommended, because of the cumulative risks of inducing serious NSAID-related adverse events.

2.4.2 Ability to Drive and Use Machines

Some patients may experience drowsiness, dizziness, vertigo insomnia or depression with the use of Naprosyn/Synflex. If patients experience these or similar undesirable effects, they should exercise caution in carrying out activities that require alertness

2.4.3 Interactions with other Medicinal Products and other Forms of Interaction

Concomitant administration of antacid or cholestyramine can delay the absorption of naproxen, but does not affect its extent.

Naprosyn and Synflex Tablets: Concomitant administration of food can delay the absorption of naproxen, but does not affect its extent.

Naproxen is highly bound to plasma albumin; it thus has a theoretical potential for interaction with other albumin-bound drugs such as coumarin-type anticoagulants, sulphonylureas, hydantoin, other NSAIDs and aspirin. Patients simultaneously receiving Naprosyn/Synflex and a hydantoin, sulphonamide or sulphonylurea should be observed for adjustment of dose if required.

Although no significant interactions have been observed in clinical studies with naproxen and coumarin-type anticoagulants, NSAIDs may enhance the effects of anti-coagulants, such as warfarin. Naproxen decreases platelet aggregation and prolongs bleeding time. This effect should be kept in mind when bleeding times are determined. Clinical pharmacodynamic data suggest that concomitant naproxen usage for more than one day consecutively may inhibit the effect of low-dose acetylsalicylic acid on platelet activity and this inhibition may persist for up to several days after stopping naproxen therapy. The clinical relevance of this interaction is not known.

Caution is advised when probenecid is administered concurrently, since increases in naproxen plasma concentrations and increased half-life of naproxen have been reported with this combination.

Caution is advised when methotrexate is administered concurrently, since naproxen and other prostaglandin synthesis-inhibiting drugs have been reported to reduce the clearance of methotrexate, and thus possibly enhance its toxicity.

Naprosyn/Synflex can reduce the anti-hypertensive effect of beta blockers, angiotensin converting enzyme inhibitors (ACE inhibitors), and angiotensin receptor blockers (ARBs).

Concomitant use of NSAIDs with ACE inhibitors or angiotensin receptor blockers may increase the risk of renal dysfunction, especially in patients with pre-existing poor renal function (see Section 2.4 Warnings and Precautions).

As with other non-steroidal anti-inflammatory drugs, Naprosyn/Synflex may inhibit the natriuretic effect of frusemide.

Inhibition of renal lithium clearance leading to increases in plasma lithium concentrations has been reported.

It is suggested that Naprosyn/Synflex therapy should be temporarily discontinued 48 hours before adrenal function tests are performed, because naproxen may artefactually interfere with some tests for 17-ketogenic steroids. Similarly, Naprosyn/Synflex therapy may interfere with some urinary assays of 5-hydroxy indoleacetic acid (SHIAA).

There is an increased risk of gastrointestinal bleeding (see section 2.4, Warnings and Precautions) when anti-platelet agents and selective serotonin reuptake inhibitors (SSRIs) are combined with NSAIDs.

2.5 Use in Special Populations

2.5.1 Pregnancy

Premature Closure of Foetal Ductus Arteriosus

Naproxen may cause premature closure of the foetal ductus arteriosus. Avoid use of naproxen in pregnant women starting at about 30 weeks of gestation (third trimester) and later. Naproxen increases the risk of premature closure of the foetal ductus arteriosus at approximately this gestational age.

Oligohydramnios/Neonatal Renal Impairment

Use of NSAIDs, including naproxen at about 20 weeks gestation or later in pregnancy may cause foetal renal dysfunction leading to oligohydramnios and, in some cases, neonatal renal impairment. These adverse outcomes are seen, on average, after days to weeks of treatment, although oligohydramnios has been infrequently reported as soon as 48 hours after NSAID initiation. Oligohydramnios is often, but not always, reversible with treatment discontinuation. Complications of prolonged oligohydramnios may, for example, include limb contractures and delayed lung maturation. In some post marketing cases of impaired neonatal renal function, invasive procedures such as exchange transfusion or dialysis were required.

If treatment is necessary between about 20 weeks and 30 weeks gestation, limit naproxen use to the lowest effective dose and shortest duration possible. Consider ultrasound monitoring of amniotic fluid if naproxen treatment extends beyond 48 hours. Discontinue naproxen if oligohydramnios occurs and follow up according to clinical practice. Therefore, Naprosyn/Synflex should not be used during pregnancy unless clearly needed.

2.5.2 Labor and Delivery

Naprosyn/Synflex are not recommended in labor and delivery because, through their prostaglandin synthesis inhibitory effect, they may adversely affect fetal circulation and inhibit uterine contractions, thus increasing the risk of uterine hemorrhage.

2.5.3 Nursing Mothers

The naproxen anion has been found in the milk of lactating women at a concentration of approximately 1% of that found in plasma. Because of the possible adverse effects of prostaglandin-inhibiting drugs on neonates, use in nursing mothers is not recommended.

2.6 Undesirable Effects

2.6.1 Post Marketing

The following adverse events have been reported with NSAIDs and with naproxen:

Gastrointestinal: Inflammation bleeding (sometimes fatal, particularly in the elderly), ulceration, perforation and obstruction of the upper or lower gastrointestinal tract. (See section 2.4, Warnings and Precautions). Oesophagitis, gastritis , pancreatitis, stomatitis. Exacerbation of ulcerative colitis and Crohn’s disease (See section 2.4 Warnings and Precautions). Heartburn, dyspepsia, abdominal pain, nausea, vomiting, diarrhoea, flatulence, constipation, haematemesis, melaena.

Infections: aseptic meningitis.

Blood and Lymphatic System Disorders:

agranulocytosis, aplastic anemia, eosinophilia, hemolytic anemia, leucopenia, thrombocytopenia.

Immune System Disorders: anaphylactoid reactions.

Metabolic and Nutrition Disorders: hyperkalemia.

Psychiatric Disorders: depression , dream abnormalities, insomnia.

Nervous System Disorders: dizziness, drowsiness, headache, lightheadedness, retrobulbar optic neuritis, convulsions, cognitive dysfunction, inability to concentrate.

Eye Disorders: visual disturbances, corneal opacity, papillitis, papilloedema.

Ear and Labyrinth Disorders: hearing impairment, hearing disturbances, tinnitus, vertigo.

Cardiac Disorders: palpitations, cardiac failure has been reported in association with NSAID treatment, congestive heart failure.

Vascular Disorders: hypertension, vasculitis. Clinical trial and epidemiological data suggest that use of coxibs and some NSAIDs (particularly at high doses and in long term treatment) may be associated with a small increased risk of arterial thrombotic events (for example myocardial infarction or stroke). Although data suggests that the use of naproxen (1000 mg/d) may be associated with a lower risk some risk cannot be excluded.

Respiratory, Thoracic and Mediastinal Disorders: dyspnea, pulmonary edema, asthma, eosinophilic pneumonitis.

Hepatobiliary Disorders: hepatitis (some cases of hepatitis have been fatal), jaundice.

Skin and Subcutaneous Tissue Disorders: ecchymoses, itching (pruritus), purpura, skin eruptions, sweating, alopecia, epidermal necrolysis, very rarely toxic epidermal necrolysis, erythema multiforme, bullous reactions, including Stevens-Johnson syndrome, Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS), erythema nodosum, fixed drug eruption, lichen planus, pustular reaction, skin rashes, SLE, urticaria, photosensitivity reactions, including rare cases resembling porphyria cutanea tarda (“pseudoporphyria”) or epidermolysis bullosa and angioneurotic edema. If skin fragility, blistering or other symptoms suggestive of pseudoporphyria occur, treatment should be discontinued and the patient monitored.

Musculoskeletal and Connective Tissue Disorders: myalgia, muscle weakness.

Renal and Urinary Disorders: hematuria, interstitial nephritis, nephrotic syndrome, renal disease, renal failure, renal papillary necrosis.

Reproductive System and Breast Disorders: female infertility.

General Disorders and Administration Site

Conditions: oedema, thirst,pyrexia (chills and fever), malaise.

Investigations: abnormal liver function tests, raised serum creatinine.

2.7 Overdose

Symptoms and signs

Significant naproxen overdosage may be characterised by dizziness, drowsiness, epigastric pain, abdominal discomfort, indigestion, nausea, transient alterations in liver function, hypoprothrombinemia, renal dysfunction, metabolic acidosis, apnoea, disorientation or vomiting.

Because Synflex may be rapidly absorbed, high and early blood levels should be anticipated. A few patients have experienced convulsions, but it is not clear whether or not these were naproxen-related. Gastrointestinal bleeding may occur. Hypertension, acute renal failure, respiratory depression and coma may occur after the ingestion of NSAIDs but are rare. Anaphylactoid reactions have been reported with therapeutic ingestion of NSAIDs, and may occur following an overdose.

Treatment

Patients should be managed by symptomatic and supportive care following NSAIDs overdose. There are no specific antidotes. Prevention of further absorption (e.g. activated charcoal) may be indicated in patients seen within 4 hours of ingestion with symptoms or following a large overdose. Forced diuresis, alkalinization of urine, hemodialysis, or hemoperfusion may not be useful due to high protein binding.

3. PHARMACOLOGICAL PROPERTIES AND EFFECTS

3.1 Pharmacodynamic Properties

Pharmacotheapeutic group: Anti-inflammatory and antirheumatic products, non-steroids; ATC code: M01AE02.

Naprosyn is a non-steroidal anti-inflammatory drug (NSAID) with analgesic, anti-inflammatory and antipyretic properties. The onset of pain relief is more rapid with Synflex than with Naprosyn, therefore Synflex is recommended for the management of acute painful conditions.

Naproxen has been shown to have striking anti-inflammatory properties when tested in human clinical studies and classical animal test systems. In addition, it has marked analgesic and antipyretic action.

3.1.1 Mechanism of Action

It exhibits its anti-inflammatory effects even in adrenalectomised animals, indicating that its action is not mediated through the pituitary axis. It inhibits synthesis of prostaglandins as with other similar agents, however, the exact mechanism of its anti-inflammatory action is not known

3.2 Pharmacokinetic Properties

3.2.1 Absorption

Naproxen and naproxen sodium are rapidly and completely absorbed from the gastrointestinal tract after oral administration. Naproxen sodium is more rapidly absorbed than naproxen. Concomitant administration of food can delay the absorption of naproxen and naproxen sodium, but does not affect its extent. Naprosyn Tablets: After administration of Naprosyn Tablets, peak plasma levels are attained in 2 to 4 hours depending on food intake. Synflex Tablets: After oral administration of Synflex Tablets, because of rapid and complete absorption, clinically significant plasma levels and pain relief are obtained in patients within 30 minutes of administration. Peak plasma levels are attained in 1-2 hours, depending on food intake. The difference in rates between the two products is due to the increased aqueous solubility of the sodium salt of naproxen. Naprosyn EC tablets: Enteric-coated naproxen dissolves primarily in the small bowel rather than the stomach, so the absorption is delayed until the stomach is emptied. When multiple doses of naproxen are administered in the EC form, the peak plasma levels, elimination half-life and the area under the plasma concentration-time curve at steady state are comparable to the non-EC form of naproxen. Naprosyn EC given as a single dose with food resulted in peak plasma levels in about 12 hours (range 4-24 hours). In fasted subjects, peak plasma levels are attained in about 4 hours following the first dose.

3.2.2 Distribution

Naproxen has a volume of distribution of 0.16 l/ kg. At therapeutic levels naproxen is greater than 99% albumin-bound. At doses of naproxen greater than 500 mg/day, there is less than proportional increase in plasma levels due to an increase in clearance caused by saturation of plasma protein binding at higher doses. However, the concentration of unbound naproxen continues to increase proportionally to dose. Steady-state plasma levels of naproxen are reached after 3-4 days. Naproxen enters synovial fluid, crosses the placenta and has been found in the milk of lactating mothers at a concentration approximately 1% of that found in plasma.

3.2.3 Metabolism

Naproxen is extensively metabolised in the liver to 6-0-desmethyl naproxen.

3.2.4 Elimination

Approximately 95% of the naproxen from any dose is excreted in the urine, primarily as naproxen (less than 1%), 6-0-desmethyl naproxen (less than 1%), or their conjugates (66-92%). The rate of excretion of metabolites and conjugates has been found to coincide closely with the rate of naproxen disappearance from the plasma. Small amounts, 3% or less, are excreted in the feces. The clearance of naproxen is approximately 0.13 ml/min/kg. The elimination half-life of naproxen is approximately 14 hours and is independent of the chemical form or the formulation.

3.2.5 Pharmacokinetics in Special Populations

Renal impairment

Given that naproxen and its metabolites are primarily excreted by the kidney, the potential exists for accumulation in the presence of renal insufficiency. Elimination of naproxen is decreased in patients with severe renal impairment. In patients who are severely renally impaired (creatinine clearance <10 ml/min), there is higher clearance of naproxen than estimated from the degree of renal impairment alone.

Children

The pharmacokinetic profile of naproxen in children aged 5-16 years is similar to that in adults although the clearance is generally higher in children than in adults Pharmacokinetic studies of naproxen were not performed in children less than 5 years of age.

3.3 Carcinogenicity

Synflex was administered with food to Sprague-Dawley rats for 24 months at doses of 8, 16 and 24 mg/kg/day. Naprosyn was not carcinogenic in rats.

3.3.2 Mutagenicity

Mutagenicity was not seen in Salmonella typhimurium (5 cell lines), Sachharomyces cerevisiae (1 cell line), and mouse lymphoma tests.

3.3.3 Impairment of Fertility

Synflex did not affect the fertility of rats when administered orally at doses of 30mg/kg/day to males and 20 mg/kg/day to females.

3.3.4 Teratogenicity

Synflex was not teratogenic when administered orally at doses of 20 mg/kg/day during organogenesis to rats and rabbits.

3.3.5 Other Perinatal/Postnatal Reproduction

Oral administration of Synflex to pregnant rats at doses of 2, 10 and 20 mg/kg/day during the third trimester of pregnancy resulted in difficult labour. These are known effects of this class of compounds and were demonstrated in pregnant rats with aspirin and indomethacin.

4. PHARMACEUTICAL PARTICULARS

4.1 Special Instructions for Use, Handling and Disposal Stability and storage condition

Do not store above 30°C, protect from light.

This medicine must not be used after the expiry date (EXP) shown on the pack.

4.2 Packs

Tablets 275 mg 100

Tablets 550 mg 100

Medicine: keep out of reach of children

CDS8.0

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Sovereign House, Miles Gray Road,
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Chennai, 600127, India